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SUMMARY

Computer Science and Engineering undergraduate who builds full-stack web applications and microcontroller-based systems, with C++ data-structure projects and daily problem-solving practice on LeetCode. Experience applying machine learning and deep learning models to classification and detection.

EDUCATION

B.Sc. in Computer Science and Engineering Jan 2024 – Present
IUBAT—International University of Business Agriculture and Technology
CGPA: 3.80 / 4.00

TECHNICAL SKILLS

Languages: C, C++, C#, PHP, JavaScript (learning), Python (learning)
Frontend: HTML, CSS, Tailwind CSS, TypeScript, React (learning)
Backend: PHP, Django (learning)
Databases: MySQL, PostgreSQL (learning)
Embedded & hardware: ESP, RFID, NFC smart cards, Gas sensor, Flame sensor, Indicators
Development tools: VS Code, Code::Blocks, Arduino IDE, GitHub, XAMPP
Computer Science Foundations: Programming, Data Structures, Algorithms, Computer Organization & Architecture, Operating Systems, Database Management Systems, Data Communications, System Analysis & Design, Data Science

PROJECTS

Dental Care Management System (DCMS), online edition | *HTML, Tailwind CSS, React, Django, JavaScript, TypeScript, Python, PostgreSQL*
- Built a bilingual (Bangla and English) clinic system with role-based access, conflict-free scheduling, billing, printable invoices, consent management and online guest entry for appointment requests.

Dental Care Management System (DCMS), local edition | *HTML, CSS, PHP, MySQL, XAMPP*
- Offline single-machine edition of DCMS with the same bilingual, role-based scheduling, billing and consent workflows.

Issue Tracker | *HTML, CSS, PHP, MySQL, XAMPP*
- Web app for reporting, categorizing and resolving issues and tasks by department, priority, status and project.

RFID IoT Server-Based Attendance System (RISBAS) | *Arduino IDE, ESP, RFID, NFC smart cards, C++, Indicators*
- Attendance recording from NFC card taps on an RFID reader wired to an ESP microcontroller, with indicator feedback.

FireGuard: An Analog Gas / Smoke and Flame Detection System | *Gas sensor, Flame sensor, Indicators*
- Sensor-based hazard detection for gas, smoke and flame that raises alerts through indicators.

University Management System (UMS) | *C++, Code::Blocks, VS Code*
- Console application with role-based dashboards for students, faculty and administrators covering academics, attendance, fees and admissions.

Smart Irrigation System | *C++, Code::Blocks, VS Code*
- Modular, object-oriented C++ simulation of soil-moisture monitoring and automatic water flow with safeguards, logging and manual override.

Smart Hospital Patient Management System | *C++, Code::Blocks, VS Code*
- Admissions, emergency prioritization, search, undo/redo and sorting built on linked lists, queues, trees and stacks.

Traffic Flow Simulation (TFS) | *C++, Code::Blocks, VS Code*
- Terminal-based multi-lane traffic simulation with priority handling, manual and automated modes, CSV logs and statistics.

RESEARCH EXPERIENCE

- Alzheimer's disease classification using deep learning: benchmarking 7 models on 1 dataset (paused).
- IoT network intrusion detection using machine learning: 4 models on 2 datasets, with cross-validation (paused).